

The Lean Startup

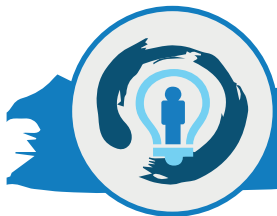
How Today's Entrepreneurs use Continuous Innovation to Create Radically Successful Businesses

BY ERIC RIES



The Big "So What"

Each time startups and innovations fail, huge amounts of money, time, passion, and skill are wasted. Yet, with the right processes, their success could be significantly improved. This book is not just about starting a new business. It's about developing new products or services successfully under conditions of great uncertainty, reducing wastage, and building an adaptive organization that can thrive in a rapidly changing world.



The 5 Lean Startup Principles

Through his own business failures, Ries found many of the traditional entrepreneurship notions to be flawed. He began to try different approaches at IMVU – his new startup that developed avatars - experimenting with the application of ideas from lean manufacturing to his entrepreneurial challenges. This eventually evolved into Lean Startup – the application of lean thinking to the process of continuous innovation.

Startups face unique challenges, which call for not just "ordinary" management, but **entrepreneurial management**. There are 5 important principles that underlie The Lean Startup method:



1. Entrepreneurs are Everywhere

A startup is "a human institution designed to create a new product or service under conditions of extreme uncertainty."

KEY QUOTES

"My mission: to improve the success rate of new innovative products worldwide. The result is the book you are reading."

"Startup success is not a consequence of good genes or being in the right place at the right time. Startup success can be engineered by following the right process, which means it can be learned, which means it can be taught."

"Entrepreneurs are everywhere and the Lean Startup approach can work in any size company, even a very large enterprise, in any sector or industry."

"We often lose sight of the fact that a startup is not just about a product, a technological breakthrough, or even a brilliant idea. A startup is greater than the sum of its parts; it is an acutely human enterprise."

It's not defined by company size, organization type (e.g. corporation, government, non-profit) nor industry.

An “intrapreneur” in a big company – e.g. a HR manager doing creative recruitment, an IT manager developing a new solution – is as much an entrepreneur as a founder of a new company. The book details several examples of the Lean Startup at work, from large companies to fresh entrepreneurial startups.



2. Entrepreneurship *is* Management

Many people think that management is dull and boring, while entrepreneurship is exciting and cool. But, **entrepreneurship is management, since a start-up is basically an institution that needs to be managed**. The Lean Startup integrates ideas from lean manufacturing (e.g. just-in-time inventory management, small batch sizes, accelerated cycle times), to the specific challenges of entrepreneurship and startups.



3. Build-Measure-Learn

Since startups' products or services are new, they don't know who their clients are. They can't predict which approach works best until they put them to the test, yet it's inefficient to test all the options via trial-and-error. **Entrepreneurs tend to either adopt a “just-do-it” approach, or suffer from analysis paralysis** and spend too long trying to “perfect” their plan or product. Either ways, by the time they realize they've created something that nobody wants, it's usually too late.

The Lean Startup's Build-Measure-Learn feedback loop allows you to learn as early as possible and make constant adjustments, so you can decide whether to change course (to “pivot”) or persevere.



4. Validated Learning

Startups need to quickly find out if customers want what

KEY QUOTES

“I believe that entrepreneurship requires a managerial discipline to harness the entrepreneurial opportunity we have been given.”

“Most tools from general management are not designed to flourish in the harsh soil of extreme uncertainty in which startups thrive.”

“For every success there are far too many failures...(these are) not just ...economic damage done to individual employees, companies, and investors; they are also a colossal waste of our civilization's most precious resource: the time, passion, and skill of its people.”

they have to offer, otherwise it's pointless to build it on time and on budget. They also need to learn how to market, sell, and operate this process effectively. Learning is essential; yet, how do you really know you've learnt the "right thing" and are creating value from it?

- The key is to determine which efforts are value-creating, which ones are wasteful, then systematically eliminate the wasteful ones.
- Yet, customers can't always tell you what they want, especially if it is a new product that they have not experienced before. Only by experimenting with different options and observing customers' actions (or inactions) can you derive true insights about their needs. **Validated learning is about measuring the results of each applied insight, so you can measure real progress from learning.**



5. Innovation Accounting

In order to measure progress towards goals and prioritise their work, startups need a different type of accounting. **Innovation accounting quantifies if our efforts are generating results, and allows us to create learning milestones to assess our progress accurately and objectively.**

In this summary, we'll explain these 5 Lean Startup principles in 3 parts: **Vision, Steer, and Accelerate.**



Part 1: Vision

Building a startup is like driving a car. Most startups have a vision or destination in mind. To achieve that vision, they employ a strategy (a business model or road map) and create products or services to achieve that vision. The key is to avoid building your strategy and products/services on untested assumptions.

Instead, build experiments to test the hypotheses relating

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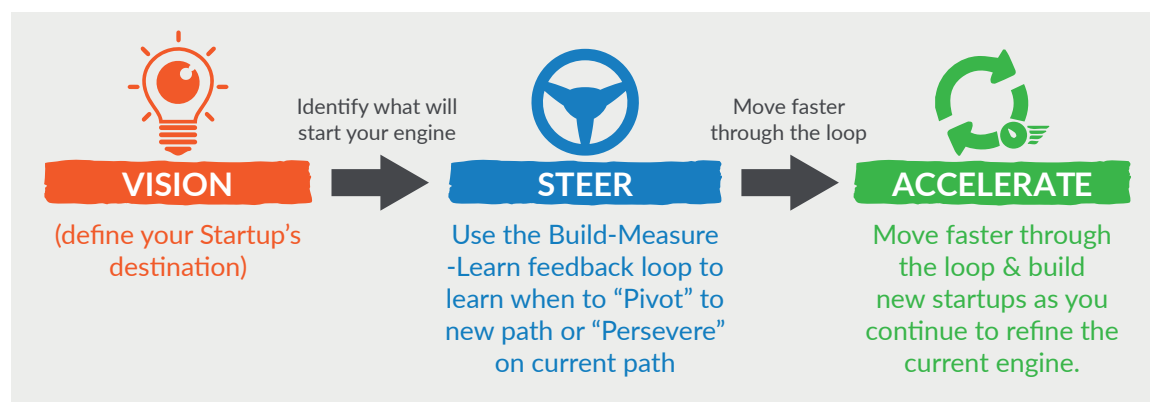
"Startups exist not just to make stuff, make money, or even serve customers. They exist to learn how to build a sustainable business."

"Customers could not tell us what they wanted...Instead, they revealed the truth through their action or inaction."

"Lean thinking defines value as providing benefit to the customer; anything else is waste."

"This is true startup productivity: systematically figuring out the right things to build."

to your vision, so you can identify and refine the key variables to start the engine. Then, you can “steer” your startup via the Build-Measure-Learn feedback loop (covered in the next segment), and use validated learning and innovation accounting to help you decide when to “pivot” or persevere. Once the engine is running, you can repeat the cycles and accelerate by progressively shifting into higher gears (covered in the last segment).



Part 2: Steer

Leap-of-Faith Assumptions



Break down your grand vision into smaller components and identify the riskiest elements, or your “leap-of-faith assumptions”. For example Nick Swinmurn had a vision for a central online site for great shoes; he tested his hypothesis that “customers were ready and willing to buy shoes online”, before building his online store. Today, Zappos is the largest global online shoe retailer. **There are 2 important assumptions you should test:**

+ Value hypothesis: Test whether a product or service truly provides value to customers who use it. Rather than ask customers (they may not be able to give an objective answer), run experiments to see how people participate, e.g. whether they sign up, how and how long they use the product etc.; and

↑ Growth hypothesis: Test how new customers will discover a product or service, and if they will actively spread the word to other potential users.

KEY QUOTES

“This is one of the most important lessons of the scientific method: if you cannot fail, you cannot learn.”

“Startups...have a true north, a destination in mind: creating a thriving and world-changing business. I call that a startup's vision.”

“The Lean Startup method... is designed to teach you how to drive a startup.”

“The goal of every startup experiment is to discover how to build a sustainable business around (its) vision.”

“In the Lean Startup model, an experiment is more than just a theoretical inquiry; it is also a first product.”

The idea is to do experiment(s) over a few weeks, rather than wait for months or years for a full plan to be developed. Such experiments can happen in parallel with strategic planning. That way:

- Once an experiment is successful, you can launch your campaign (and get even more inputs for refinements); and
- By the time the product is ready for mass-distribution, it will already have established customers, solve real problems, with detailed specifications for what customers want

Build (Minimum Viable Product)



Once we have defined our leap-of-faith assumptions, we are ready to test them with a Minimum Viable Product (MVP). This is when we enter the “Build” phase.

The MVP need not be a prototype or a smallest version of the product. It's simply a tool to test our hypotheses and complete the Build-Measure-Learn loop, with the least possible time and effort. We want to understand our potential customers and their problems, confirm that they have a real problem worth solving, and will buy our solution before we go about building it. [Here are some examples of MVPs:](#)

- Video. Dropbox realized that building a working prototype would take a lot of time and resources, and they needed inputs about a problem that people didn't yet know they had. So, they made a 3 minute video to explain to technology early adopters how Dropbox would work. Through viewers' signups, they validated that people wanted the product that they were about to built.
- Road trip. When Toyota wanted to create a new model of its minivan Sienna for North America, chief engineer Yuji Yokoya started with a test of his critical assumptions of what consumers wanted in a minivan. He made a road trip across 50 U.S. states, 13 Canadian territories, and all parts of Mexico, driving a then-model of Sienna to talk to and observe real customers.

KEY QUOTES

“What differentiates the success stories from the failures is that the successful entrepreneurs had the foresight, the ability, and the tools to discover which parts of their plans were working brilliantly and which were misguided, and adapt their strategies accordingly.”

“Unlike a prototype or concept test, an MVP is designed not just to answer product design or technical questions. Its goal is to test fundamental business hypotheses.”

“Metrics are people too... Successful sales models depend on breaking down the monolithic view of organizations into the disparate people that make them up.”

“Deciding exactly how complex an MVP needs to be cannot be done formulaically. It requires judgment.”

- Single deal. Groupon started out as a “collective activism platform” called The Point, with the aim of bringing people together to solve problems they couldn’t solve on their own. It didn’t work. Then, in 2008, they tried a new MVP – this time, 20 people bought a 2-for-1 pizza in a restaurant on the ground floor of their office block, and their engine started from there.

Here are some important tips for MVP / hypothesis testing:

 Start with early adopters (people who most badly need the solutions). This group is most likely to give feedback, and be more forgiving of mistakes.

 Building an MVP comes with some legal, branding, competitive and morale risks, especially since the quality is usually bad and there may be many failed iterations. Consider launching the MVP under a different brand name, so you can experiment under the radar before doing a public marketing launch with a proven product and real customers. You can also secure prior commitment that no matter what happens during MVP testing, the team will not give up hope or abandon the project.

Measure



In this phase, our key challenge is determining whether our product development efforts are making real progress. We need to know that we’re learning the “right” things, and if our changes are delivering results. This is crucial to minimize wastage, and also to assure investors and stakeholders that we are making progress.

We can address this using innovation accounting, with 3 milestones:

 Establish the baseline. Where are you now, and how are you currently performing?

KEY QUOTES

“The lesson of the MVP is that any additional work beyond what was required to start learning is waste, no matter how important it might have seemed at the time.”

“Only 5 percent of entrepreneurship is the big idea, the business model, the whiteboard strategizing, and the splitting up of the spoils. The other 95 percent is the gritty work that is measured by innovation accounting: product prioritization decisions, deciding which customers to target or listen to, and having the courage to subject a grand vision to constant testing and feedback.”



Tune the engine. Initiate the experiment to test your value or growth hypothesis. e.g. improving your product design to increase new customer activation / usage of your product.



Pivot or persevere. Based on the results or data, you can persevere with the product/features/process you now know to be correct, or pivot and test another assumption. Although a pivot means re-establishing a new baseline (and tuning the engine from there), you'll know it's successful if the engine-tuning efforts are more productive after the pivot.

It's important to use **Actionable Metrics, not "Vanity" Metrics.** Gross numbers like total revenue and total number of customers can be misleading. Instead, **look at actionable metrics that measure actual behaviors behind desired performance. You can use a split-test experiment,** i.e. offer 2 different versions of a product to customers at the same time, and observe if there are any differences in behaviours (e.g. signup rate, length of use, purchases). You may discover that some features (which you believed were important) have no impact on customer behaviour.

For such metrics to work, they must be:



Actionable: They must display clear cause and effect, so people know what action to take.



Accessible: First, make the reports as simple as possible so everyone understands them (e.g. instead of "a website hit", use "a person visiting the website"). Second make the reports widely accessible e.g. mail the document to every employee of the company, or make it available on your intranet.



Auditable: Ensure the data is credible to employees, e.g. the data can be validated with real customers, and/or the reports are drawn directly from master data (rather than intermediate systems) to reduce opportunities for error.

KEY QUOTES

"When cause and effect is clearly understood, people are better able to learn from their actions. Human beings are innately talented learners when given a clear and objective assessment."

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Learn: (Pivot or Persevere)



Everything discussed so far is to equip us to answer this question: Are we making enough progress to believe our original strategic hypothesis is correct (hence we persevere), or should we make a major change (and pivot)? Unfortunately, this is not an easy decision, and there's no way to replace the human elements (of vision, intuition, and judgment) using scientific analysis. **Let's take a closer look at the pivot concept.**

 If you see a decreasing effectiveness of your product experiments, or feel product development could be more productive, it's a tell-tale sign of the need for a pivot. A pivot is a structured course correction, designed to test a new hypothesis about your product, strategy, and engine of growth. Each pivot leads to new opportunities, experiments, and the cycle repeats – establish the baseline, tune the engine, make a decision to pivot or persevere.

 A startup's "runway" is the number of pivots it can still make, or the number of chances it has left to make a fundamental change and take off. From this perspective, a startup's goal must be to get to the pivot asap, by moving through the feedback loop asap.

 However, pivots can be hard to make. First, many businesses use *broad, vanity metrics* that blind them to the need for necessary change. Second, entrepreneurs with an *unclear hypothesis* cannot establish if an experiment has failed. Third, it's scary to acknowledge failure, i.e. *pivots require courage*.

 Hence, the decision to pivot must be addressed in a structured, objective way. One solution is to schedule a regular "pivot or persevere" meeting in advance. At the meetings, the *product development team* presents detailed reports on the product optimization results over time, and the *business leadership team* presents detailed accounts of conversations with current and potential customers.

KEY QUOTES

"Startup productivity...is about aligning our efforts with a business and product that are working to create value and drive growth."

"Entrepreneurs need to face their fears and be willing to fail, often in a public way."

A pivot is not just any system of change: it must test a new, fundamental hypothesis. Here are several pivots examples:

 Zoom-in pivot: What was previously considered a single feature in a product becomes the entire product.

 Zoom-out pivot: Conversely, what was previously considered the whole product becomes only a feature of a much larger product.

 Customer segment pivot: The product solves a real problem for real customers, but not the original customers you intended. At some point in a startup, you may also want to go beyond early adopters to start selling to mainstream customers. However, the latter have different requirements and are much more demanding, and a pivot will be needed to re-examine your assumptions.

 Customer needs pivot: After knowing customers extremely well, it becomes clear that the problem you were trying to solve isn't very important to them. However, you discover other related problems that are important and can be solved by your team.

 Platform pivot: The solution is changed from an application to a platform, or vice versa.

 Business architecture pivot: This involves moving from a high margin, low volume model to a low margin, high volume model, or vice-versa. An example is to pivot from a Business-to-Business to Business-to-Consumer model.

 Value capture pivot: This involves a change in the way the company captures and/or monetizes its products and services.

 Engine of growth pivot: This involves a change in growth strategy (see next segment for the 3 primary engines of growth).

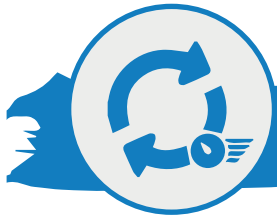
 Channel pivot: The company changes the way in which it delivers its product to customers e.g. pivot from retail stores to direct delivery.

KEY QUOTES

"The critical skill for managers today is to match those theories to their present situation so that they apply the right advice at the right time."



Technology pivot: The company may discover how to achieve the same solution using a different technology, though this is more likely to occur in established businesses.



Part 3: Accelerate

So far, we've looked at breaking down a startup idea to its leaps-of-faith assumption, testing it with an MVP, using innovation accounting and actionable metrics to evaluate the results, and making the decision to pivot or persevere. All these set the foundation for the last part - acceleration.

Small-Batch Approach



By now, we've established the importance of getting through the feedback loop as quickly as possible. A "small-batch approach" allows us to complete one product at a time quickly, whereas a "large-batch approach" delivers all the products together at the end.

For example, instead of producing thousands of parts at a go, Toyota produced a range of parts in small batches. This allows Toyota to produce entire vehicles in small, varied batches, to better serve smaller, more fragmented markets in addition to larger markets. Any worker can also stop the production line if they spot a defect - problems are fixed on the spot (rather than compounded through mass production).

Pull, not Push (Just-in-time management)



Another lesson from lean production is the concept of "pull" in just-in-time production. In a Lean Startup, product development should only respond to "pull requests" from an experiment to be run. Once a hypothesis is formed, the product development team engineers the design and completes the experiment as quickly as possible, using the smallest possible batch size. Any other work or features "pushed" by the product development team is waste.

KEY QUOTES

"The essential lesson is... by reducing batch size, we can get through the Build-Measure-Learn feedback loop more quickly than our competitors can. The ability to learn faster from customers is the essential competitive advantage that startups must possess."

Growth Strategies



Sustainable growth for any startup is characterized by new customers who come from the actions of past customers : (a) Word of mouth, (b) as a side effect of product use, e.g. status symbols and networks, (c) through funded advertising and (d) through repeat purchase or use.

These 4 sources of sustainable growth spin the feedback loop like a combustion engine. Ries calls these the **3 Engines of Growth**:



The sticky engine of growth: Companies with this growth strategy rely on a high customer retention rate, e.g. telco service providers. So long as the rate of new customer acquisition exceeds the attrition rate or “churn rate”, the product will grow. **Such companies must hence track their churn rate closely.**



The viral engine of growth: Here, awareness of the product spreads rapidly from person to person similar to how a virus becomes an epidemic. The “viral coefficient” quantifies the virality of a product. A viral coefficient of 0.1 means that only 1 in every 10 customers will bring a friend along, and this loop will eventually fizzle out. However, **a viral loop with a coefficient that's greater than 1.0 will grow exponentially**, as each customer will bring, on average, more than 1 other person.



The paid engine of growth, which relies on paying to acquire customers e.g. through advertisements. A company can increase this rate of growth **either by increasing the revenue from each customer, or reducing the cost of acquiring a new customer.**

Startups should focus on just 1 engine of growth, specializing in everything that is required to make it work, before considering a pivot.

KEY QUOTES

“The Lean Startup works only if we are able to build an organization as adaptable and fast as the challenges it faces. This requires tackling the human challenges inherent in this new way of working.”

The Adaptive Organization



Any engine of growth will eventually slow or stop. Hence, a sustainable business must concurrently tune its engine of growth and develop new sources of growth. To do this, we need to create an adaptive organization with the right structure, culture, discipline, and can automatically adjust its processes/ performance to current conditions.

Startups that move too fast may neglect problems which compound over time. To regulate themselves, lean startups can use the “Five Whys” system - slow down and invest just the right amount of resources to tackle the problems, before speeding up again:

- This is the investigative method of asking the question “Why?” 5x to understand the root causes behind the obvious symptoms. E.g. Why did the customer not collect the purchase (because the coupon was not delivered)... Why was the coupon not delivered, and so on.
- Make proportional investments for each of the 5 “Why” levels, focusing resources on the biggest, most painful problems.
- For the system to work, these 2 rules must be followed:
 1. Be tolerant of all mistakes the first time.
 2. Never allow the same mistake to be made twice.
- The book also covers useful tips like how to avoid the 5 blames, face unpleasant truths, start small, and appoint a Five Whys Master.

Ongoing Innovation



Large companies can retain the capacity for innovation, creativity, and growth by adopting “portfolio thinking”, and structuring themselves to nurture disruptive innovation. To set up an innovative company:

- Give startup teams complete autonomy to develop and market new products within their mandate (just like how a startup would). Create cross-functional teams with full-

KEY QUOTES

“The Lean Startup works only if we are able to build an organization as adaptable and fast as the challenges it faces. This requires tackling the human challenges inherent in this new way of working.”

“What really matters is not the raw numbers or vanity metrics but the direction and degree of progress.”

“To work, startups require built-in speed regulators that help teams find their optimal pace of work.”

“Both successful startups and established companies alike must learn to juggle multiple kinds of work at the same time, pursuing operational excellence and disruptive innovation. This requires a new kind of portfolio thinking.”

time reps from all departments involved in the creation or launch of the products.

- Use resources wisely. Startups may need much less capital than big companies, but they must be accountable for every dollar.
- Give entrepreneurs a personal stake in the outcome of their creations – this could be through stock options, equity ownership, long-term performance incentives, or simply acknowledgement.
- Create a platform for experimentation, with clear ground rules for the startup teams to operate:
 - (i) To protect the parent organization, *create a “sandbox”* that allows the startup team to work freely, yet contain the impact of the new innovations. e.g. it could be specific pages/ user flows for an online service, or a single store for a retail chain.
 - (ii) *Hold teams accountable* using innovation accounting. Their goal is build the model based on customer archetypes, launch an MVP to establish a baseline, and tune the engine toward the vision.
 - (iii) *Integrate successful innovations back into the parent organization.* The entrepreneurs could move on to next startups, or be given an option to follow their project back to the parent organization.

KEY QUOTES

“A startup’s work is never done...even established companies must struggle to find new sources of growth through disruptive innovation.”

“Keep in mind that structure is merely a prerequisite—it does not guarantee success. But getting the structure wrong can lead to almost certain failure.”

“The Lean Startup is not a collection of individual tactics. It is a principled approach to new product development. The only way to make sense of its recommendations is to understand the underlying principles that make them work.”